24.04.2026

SCHRÖDINGER'S FLOW

challenge 2: in situ live/dead staining of bacterial cultures

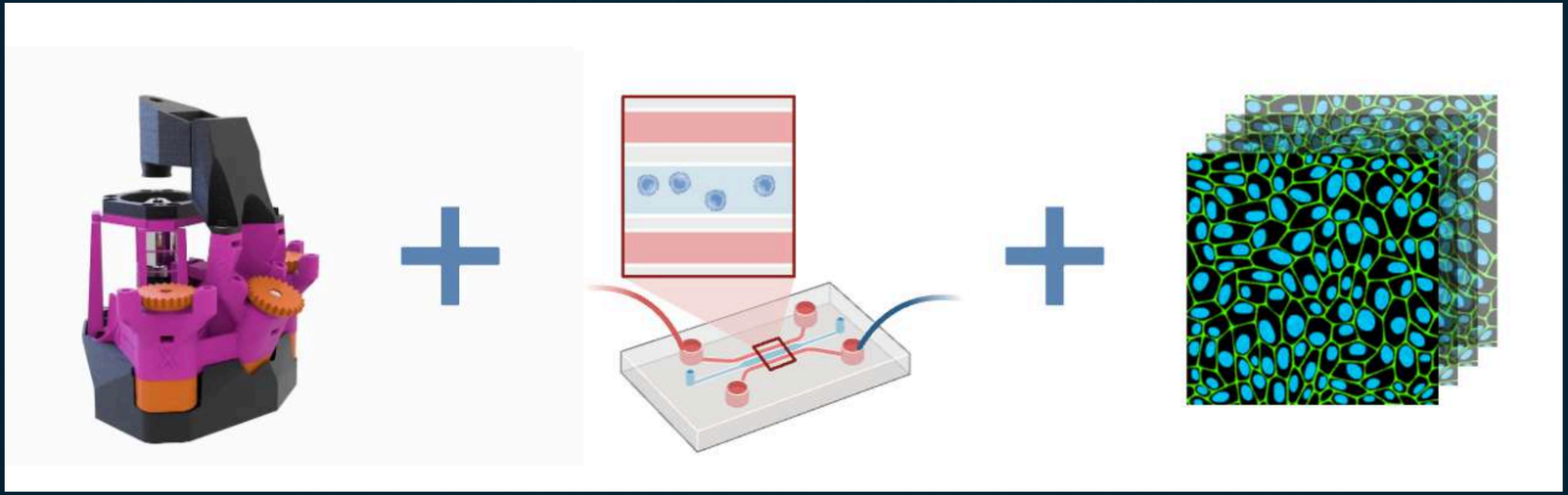
Yatharth, Mariana, Luca, Alexandra

HACK YOUR MICROSCOPE

OBJECTIVE

automated microfluidic staining and imaging system

OUR “SIMPLE” DESIGN

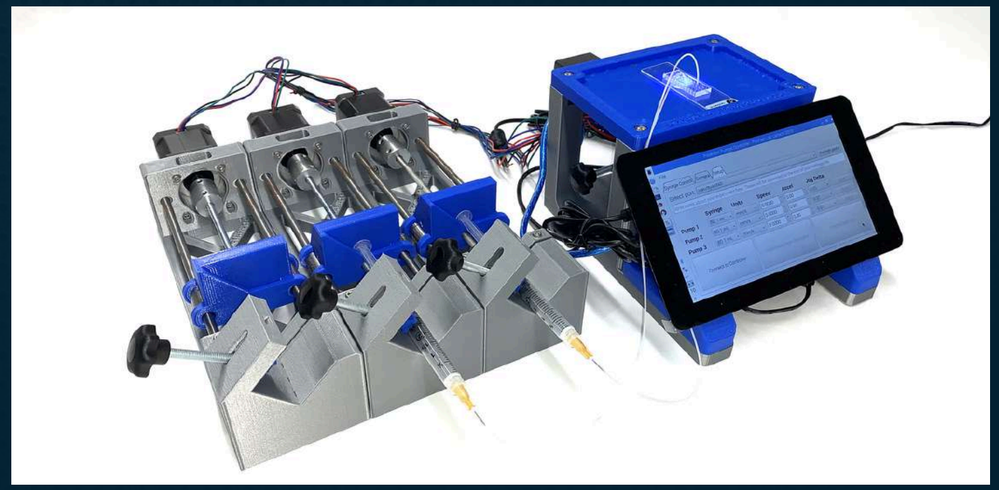


MICROSCOPE

OPENFLEXURE
WITH TWO FLUORESCENCE CHANNELS
AND A TWO CAMERA SYSTEM

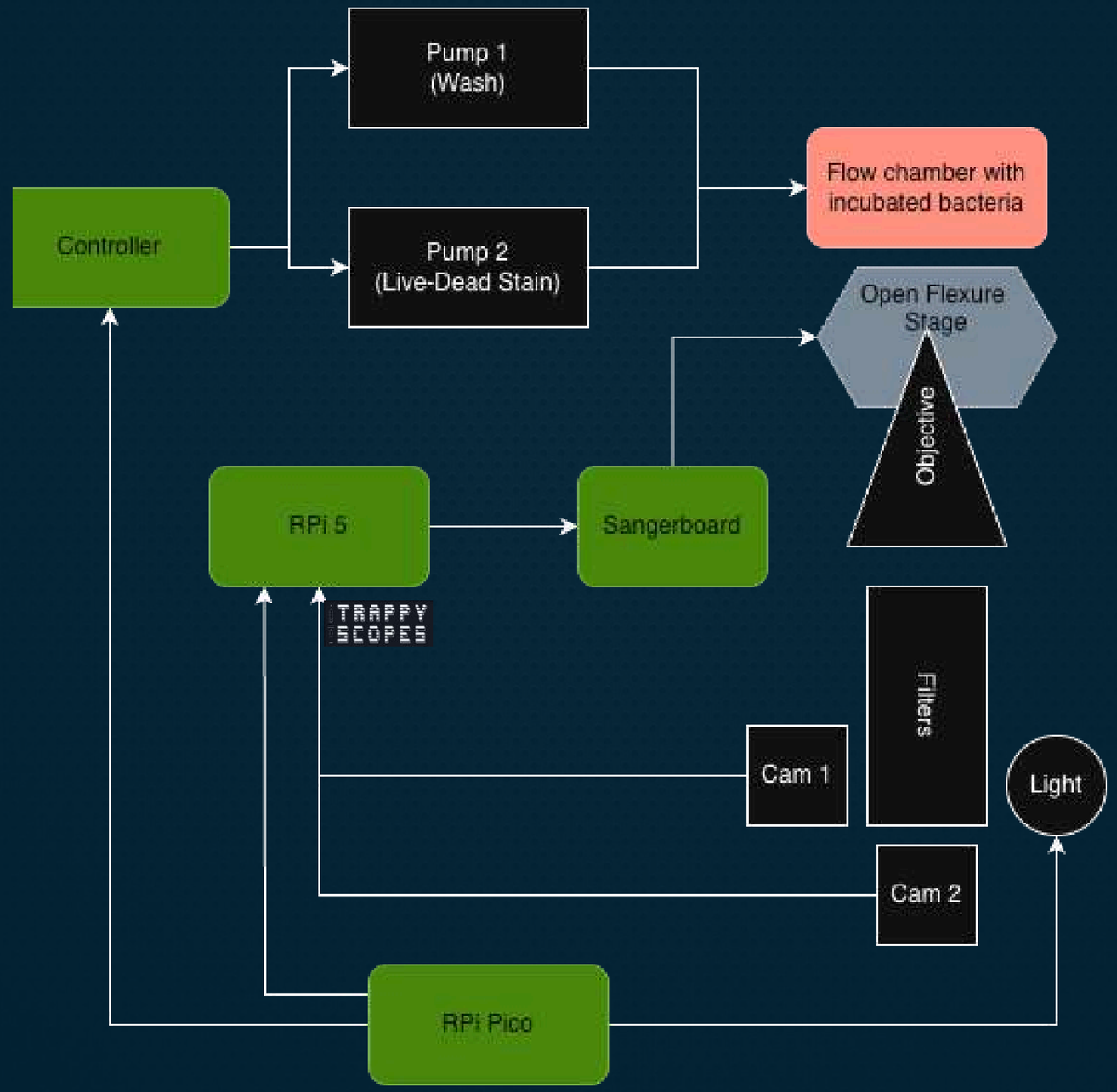
MICROFLUIDICS

TWO CHANNEL POSEIDON PUMP SYSTEM
WITH AN IBIDI FLOW CHANNEL



SHRÖDINGER

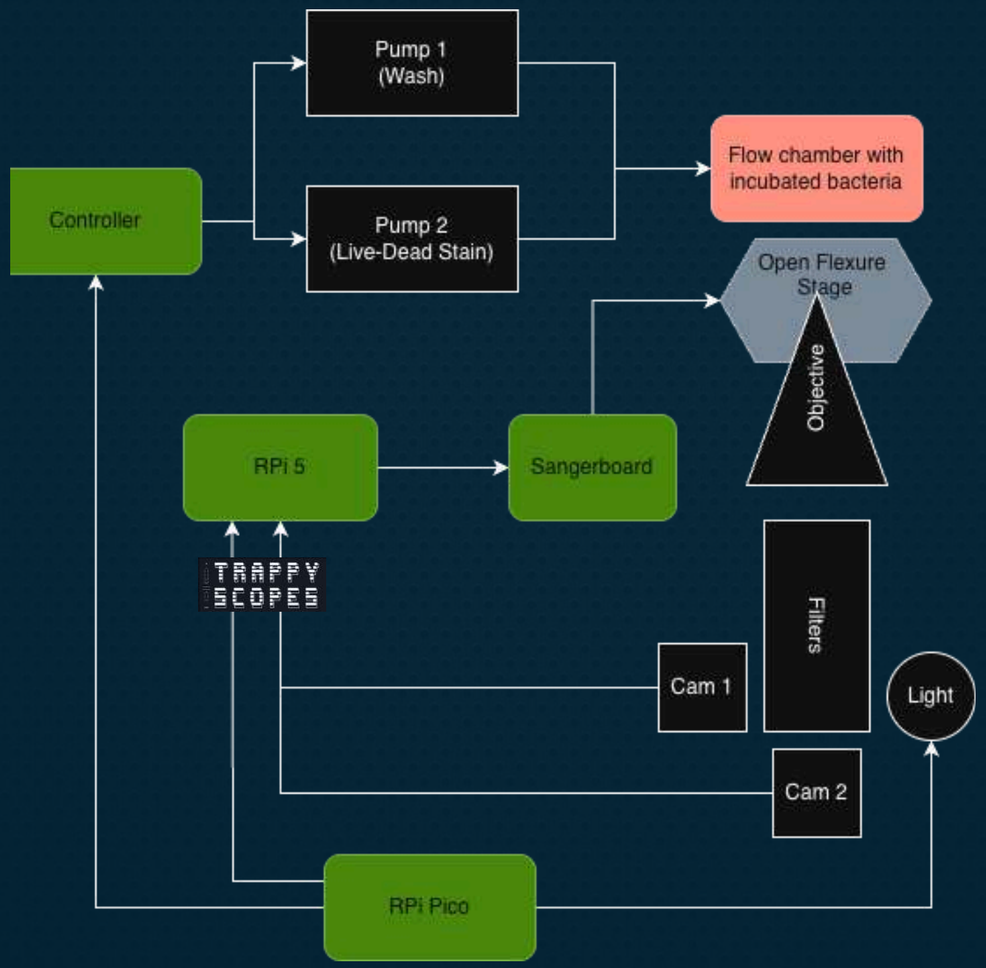
FLOW

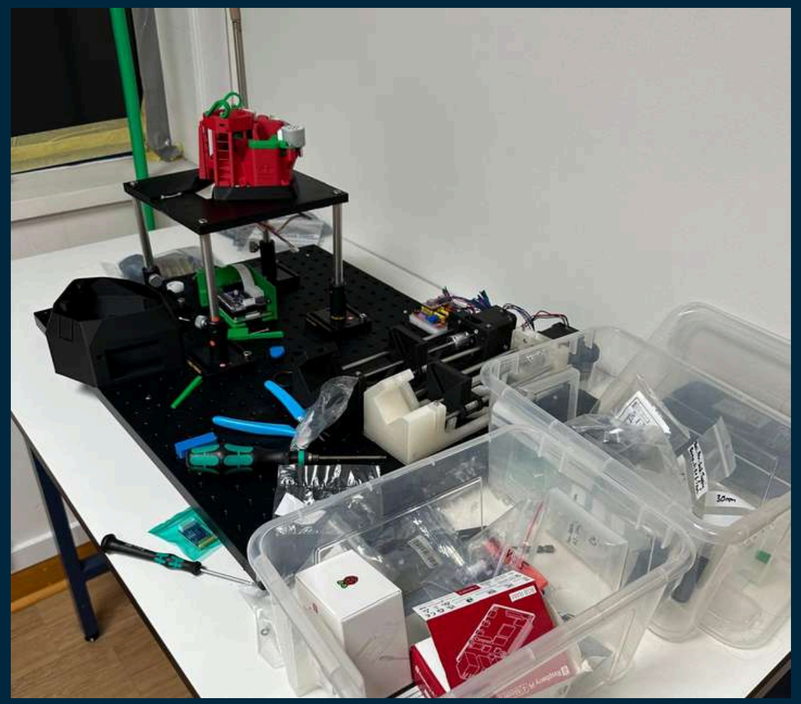


SHRÖDINGER

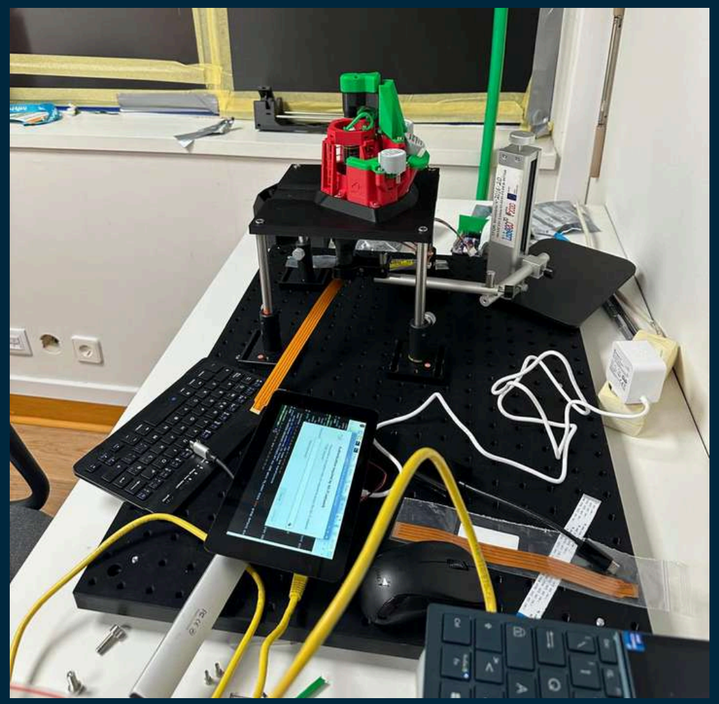
FLOW

BIOLOGY

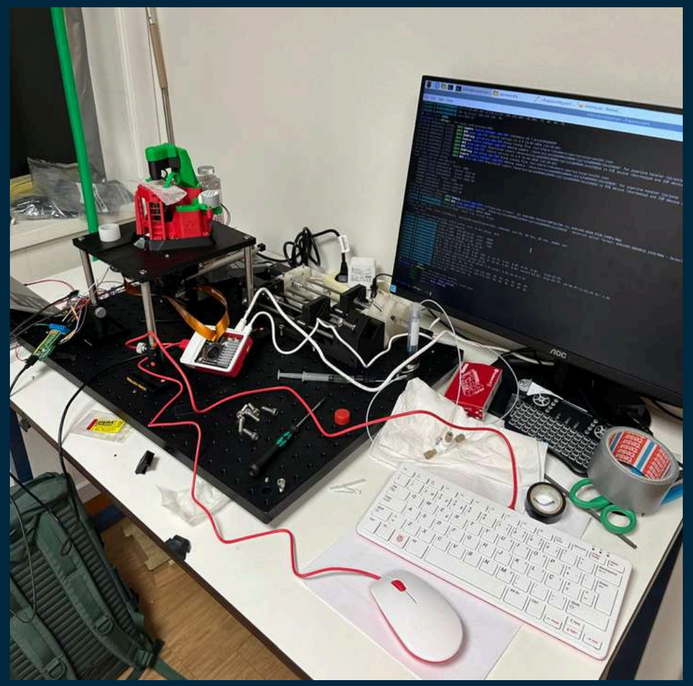




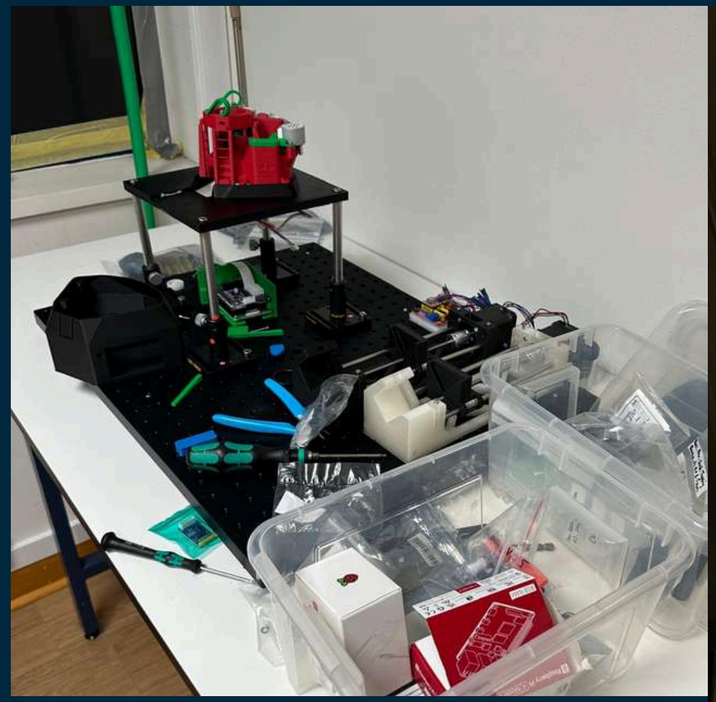
YESTERDAY @ 22:00



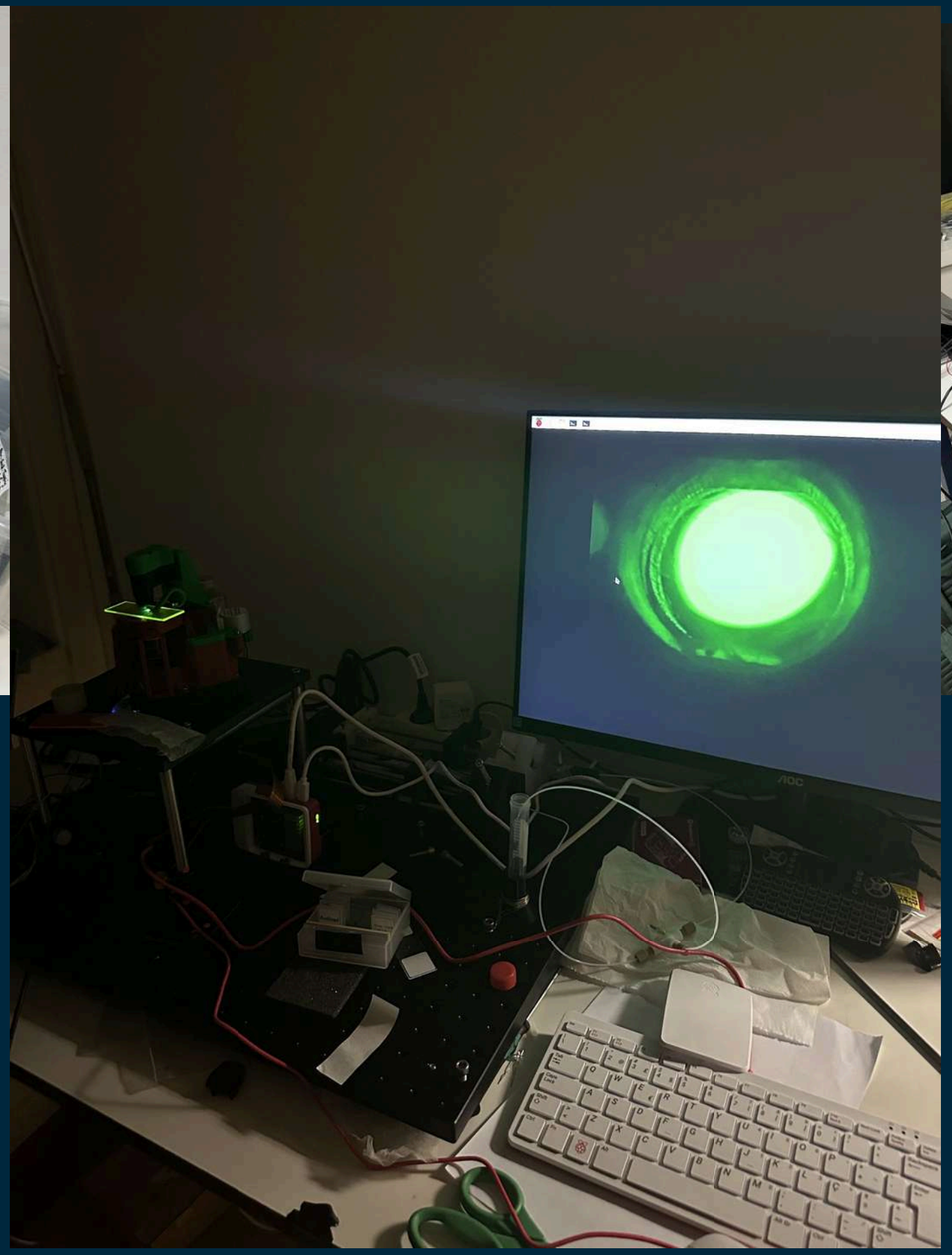
TODAY @ 16:00

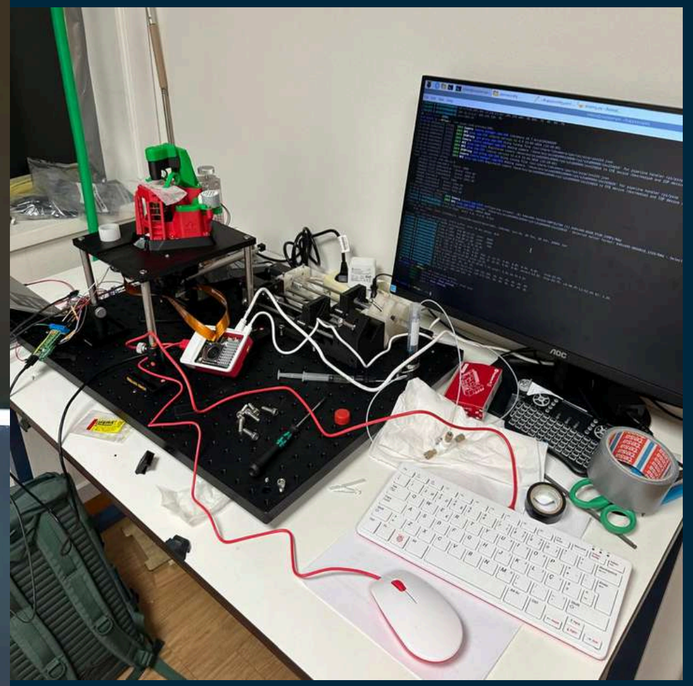


TODAY @ 19:30



YESTERDAY @ 22:00

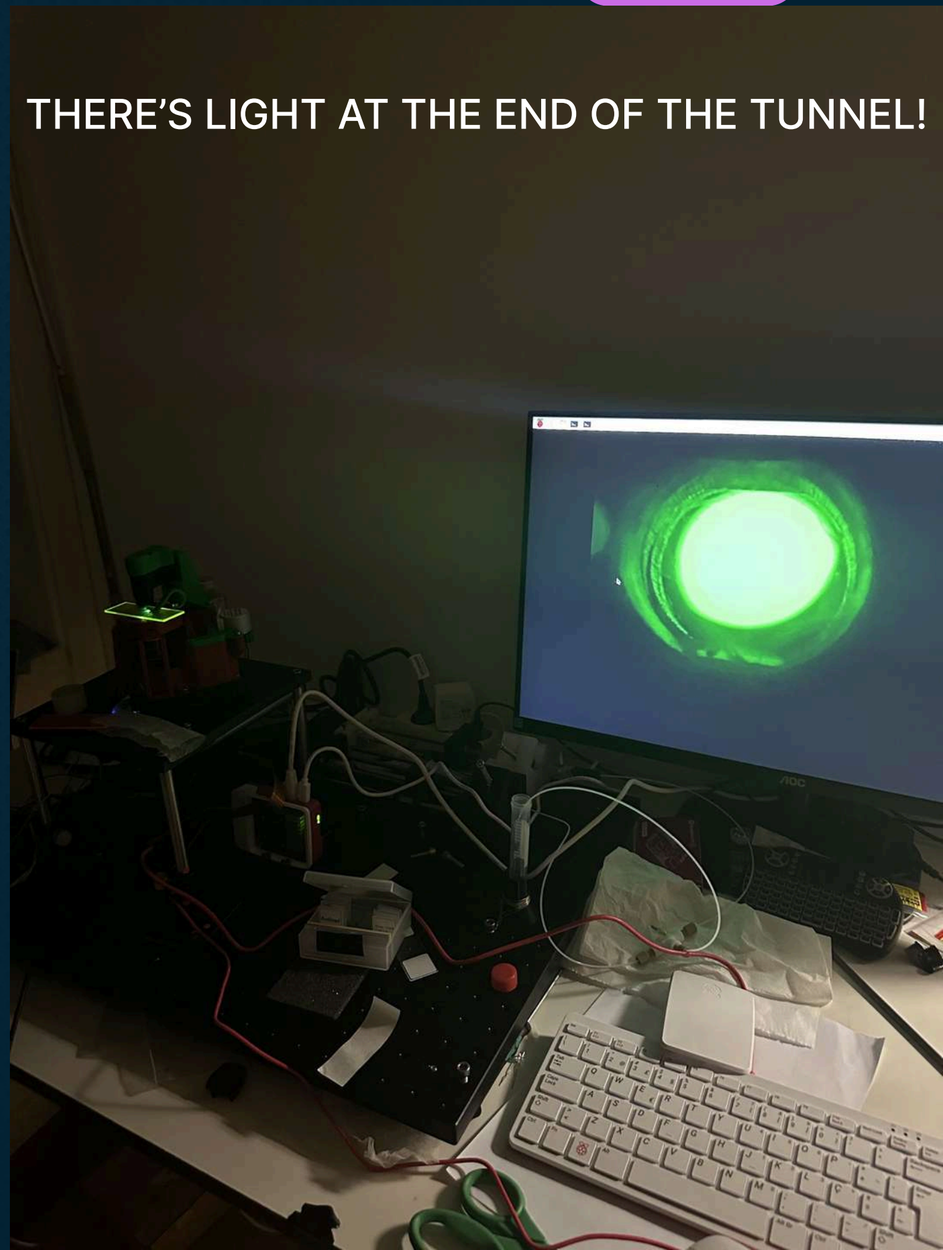




TODAY @ 19:30

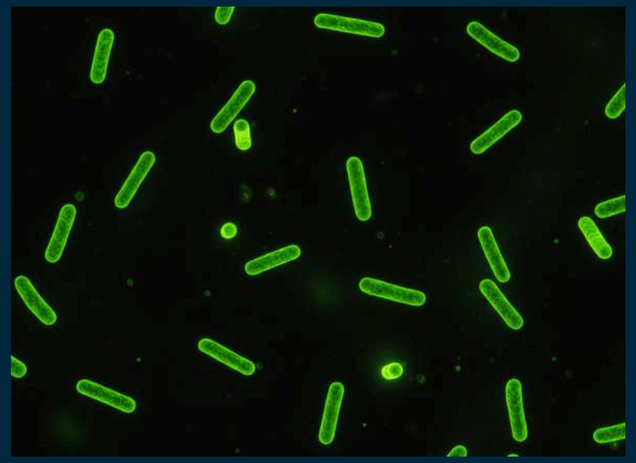
TODAY @ 22:10

SETUP
IT WORKS (MOSTLY)

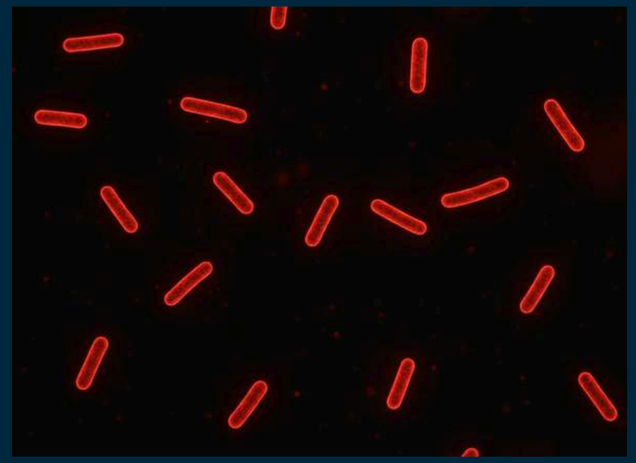


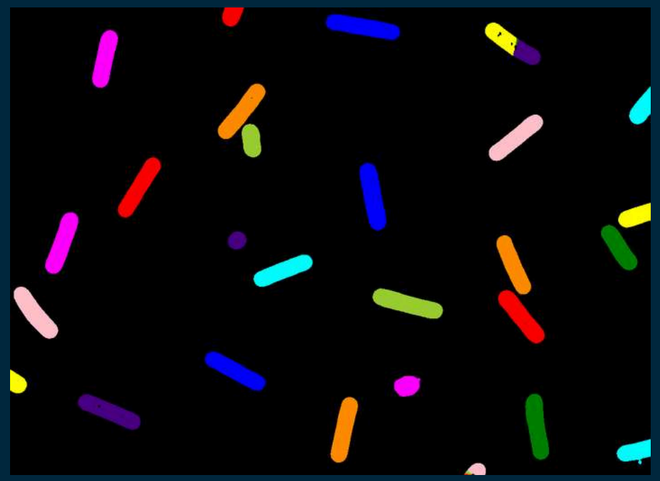
SEGMENTATION

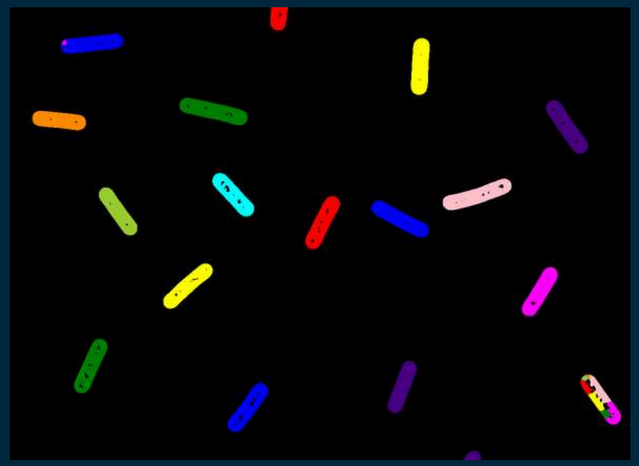
LIVE CELLS

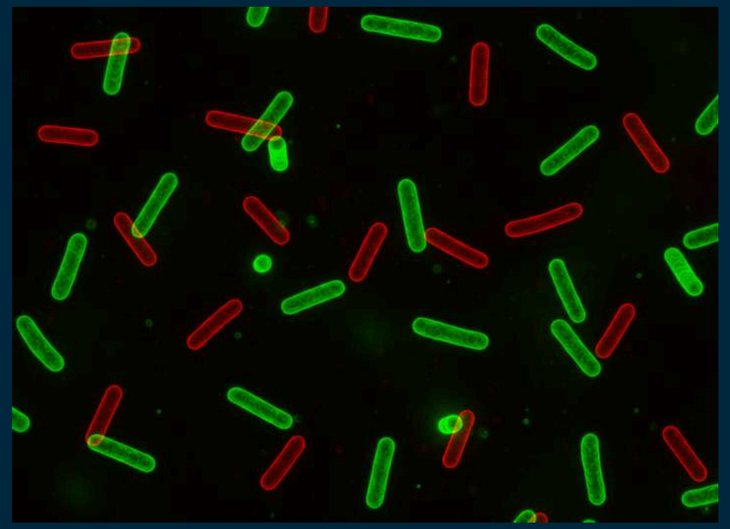


DEAD CELLS









DEAD OR ALIVE?

REAL OR AI?

```
Total cells: 30
Dead cells: 25
Live cells: 5
Live/Dead ratio: 0.2
```

FUTURE:

- (LITTLE) OPTIMIZATION
- ADD A LENS (MAYBE)
- BUY MORE TAPE

❤ Thank You!